Exercise 3.2

1. Find the angle between the lines whose slopes are $\frac{1}{2}$ and 3
2. Find the distance of the point (4,1) from the line $3x - 4y + 12 = 0$
3. Show that the straight lines $x + y - 4 = 0$, $3x + 2 = 0$ and $3x - 3y + 16 = 0$ are concurrent.
4. Find the value of 'a' for which the straight lines $3x + 4y = 13; 2x - 7y = -1$ and $ax - y - 14 = 0$ are concurrent.
5. A manufacturer produces 80 TV sets at a cost of ₹2,20,000 and 125 TV sets at a cost of ₹2,87,500. Assuming the cost curve to be linear, find the linear expression of the given information. Also estimate the cost of 95 TV sets.

3.3 Pair of Straight Lines

3.3.1 Combined equation of the pair of straight lines

Let us consider the two individual equations of straight lines

$l_1x + m_1y + n_1 = 0$ and

$l_2x + m_2y + n_2 = 0$

Then their combined equation is

$$(l_1x + m_1y + n_1)(l_2x + m_2y + n_2) = 0$$

$$l_1l_2x^2 + (l_1m_2 + l_2m_1)xy + m_1m_2y^2 + (l_1n_2 + l_2n_1)x + (m_1n_2 + m_2n_1)y + n_1n_2 = 0$$

Hence the general equation of pair of straight lines can be taken as

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

where a, b, c, f, g and h are all constants.

3.3.2 Pair of straight lines passing through the origin

The homogeneous equation

$$ax^2 + 2hxy + by^2 = 0 \quad \ldots (1)$$

of second degree in x and y represents a pair of straight lines passing through the origin.

Let $y = m_1x$ and $y = m_2x$ be two straight lines passing through the origin.

Then their combined equation is

$$(y - m_1x)(y - m_2x) = 0$$

$$\Rightarrow \quad m_1m_2x^2 - (m_1 + m_2)xy + y^2 = 0 \quad \ldots (2)$$

(1) and (2) represent the same pair of straight lines

$$\therefore \quad \frac{a}{m_1m_2} = \frac{2h}{-(m_1 + m_2)} = \frac{b}{1}$$

$$\Rightarrow m_1m_2 = \frac{a}{b} \text{ and } m_1 + m_2 = -\frac{2h}{b}.$$

i.e., product of the slopes = $\frac{a}{b}$ and sum of the slopes = $-\frac{2h}{b}$

3.3.3 Angle between pair of straight lines passing through the origin

The equation of the pair of straight lines passing through the origin is

$$ax^2 + 2hxy + by^2 = 0$$

Let m_1 and m_2 be the slopes of above lines.

Here $m_1 + m_2 = -\frac{2h}{b}$ and $m_1m_2 = \frac{a}{b}$.

Let θ be the angle between the pair of straight lines.

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